jax-experiments writeups Milestone M2

# M2 — enwik8 and the copy task in JAX (R2, R3)

A writeup in the jax-experiments series, built on 2026-10-30 from `docs/m2-enwik8-and-copy.md`. Its data, scripts and the work order (`PLAN.md`) are in the same repository.

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### At a glance

**Status:** done (R2 and R3) ·
**Hardware:** Kaggle T4 ×2 (sm_75, one job per GPU), JAX 0.11.1 on the Kaggle image. R3's fused-attention transformer ran on a Colab T4 (same JAX, Flax pinned to Kaggle's 0.12.10), because its Kaggle kernel sat queued for over 20 minutes. Parity checks on CPU (4 cores) ·
**Data:** `data/r2-summary.json` (every number below), `data/r2/` (one JSON and log per run), `data/r2-calibrate/`, `data/r2-parity.json`; ml-ladder's references from its `docs/writeups/data/c-gpu*.json` ·
**Reproduce:** `tools/kaggle/package_source.sh`, then `uv run python tools/kaggle/push_r2.py <preset> --push` for `calibrate`, `calibrate-fused`, `r2`, `r2-fused`, `r3`, `r3-s1`, `r3-s1b` (the `r3-fused` preset ran on Colab instead, as `run.py --studies copy --copy-models transformer-4M --seeds 0 1 2 --amp --attention fused`); then `uv run python experiments/r2_enwik8/analyze.py`. CPU parity: `uv run --extra torch python experiments/r2_enwik8/parity.py`

## Question

Do ml-ladder's Part C numbers reproduce in a second framework? On the same GPU, with the same protocol, does a JAX port land on the PyTorch numbers? And how do

XLA's fusion and `lax.scan` change the cost?

The arms are PLAN.md's:

- the 25M RoPE transformer at T=1024 (target 1.325 bits/byte);
- the 2×592 GRU written as `lax.scan` (target 1.598; cuDNN ran 182k tokens/s);
- the 4M transformer at lr 2e-3 (target 1.544);
- R3, the copy task: the transformer should copy 512 symbols back, and the LSTM should fail at 128.

## The idea

**Make the comparison as tight as a comparison across frameworks can be.** Each run here starts from *ml-ladder's own initial weights*. Its PyTorch model is built with the same seed and ported in (`--init torch`). Training then sees *the same batches*, drawn by the same NumPy RNG calls. So seed $s$ here is the same experiment as ml-ladder's seed $s$, and the two differ only in the arithmetic: XLA against cuDNN and cuBLAS, and the fp16 handling. Comparing the runs seed by seed is then a stronger test than comparing medians.

**Check that equality on CPU before spending GPU hours.** Sixty steps of ml-ladder's own `train()` and of `jaxexp.train` must agree to round-off, before any T4 is involved.

**Measure on the GPU before sizing the runs.** A 5-minute calibration kernel measures compiled memory and throughput on the T4. CPU numbers turned out not to transfer.

## The math

The protocol is ml-ladder's `c_context_scaling.py`:

- **Optimiser:** AdamW with $\beta = (0.9, 0.95)$, $\epsilon = 10^{-8}$, and global-norm clipping at 1.0.
- **Learning rate:** linear warmup over the first 5% of steps, then cosine decay to 10% of the peak.
- **Budget:** 12,000 steps of 16 × 1024 bytes (196.6M tokens), then per-position cross-entropy on 262,144 validation bytes.
- **Mixed precision.** PyTorch's fp16 autocast has no JAX counterpart, so the split is explicit:
  - in fp16: matmul inputs (the `compute_dtype`);
  - in fp32: LayerNorm, softmax, RoPE, the residual stream, the gates of the recurrences, and the loss.
- **Dynamic loss scaling, as GradScaler does it:**
  - start at $2^{16}$;
  - multiply by 2 after 2,000 finite steps;
  - on any non-finite gradient, skip the step and multiply by 0.5.

**Attention.** ml-ladder routes every transformer through PyTorch's SDPA, which computes $\text{softmax}(QK^\top/\sqrt{d} + M)V$ with fp16 inputs and fp32 accumulation of the scores. Two JAX paths compute the same function:

- `jax.nn.dot_product_attention` ("fused") accumulates the scores in fp32;

- the explicit einsum path *did not* until this milestone. With fp16 inputs, its score matrix came out of the matmul in fp16 and was rounded before the fp32 softmax. The absolute error is about $|s| \cdot 2^{-11}$, which is negligible for diffuse attention and about 0.02 at $|s| \approx 50$.

## What we measure and why

- **Bits/byte, per seed and as median (min–max).** Paired with ml-ladder's same-seed run, since both start from identical weights and batches.
- **Loss by position.** The median over seeds at each of the 1,024 positions, against ml-ladder's recorded curve. A framework difference in masking or positions would show up here first.
- **Cost:** steady-state tokens per second, compile time, and peak device memory, against the recorded PyTorch runs on the same T4 type.
- **fp16 health:** skipped steps and the final loss scale.
- **R3:** second-half accuracy on 512 test sequences, for n = 32, 128 and 512, seeds 0–2. A failed n=512 seed got two rounds of controls (below).

## What I built

- **`jaxexp.text`:**
  - enwik8, SHA-256 checked, with ml-ladder's split (90M train, the next 5M validation);
  - its batch sampler, byte for byte;
  - validation windows;
  - the copy task.
- **`jaxexp.recurrent`:** GRU, LSTM and Elman RNN in PyTorch's equations, as `lax.scan`. Parameters are keyed by `state_dict` names, so ml-ladder's initial weights load without renaming. The input projection runs once over the whole sequence; only the recurrent matmul is inside the scan.
- **`jaxexp.train`:**
  - ml-ladder's schedule (verbatim);
  - AdamW with clipping (`optax`, the learning rate injected per step);
  - fp16 compute with loss scaling;
  - a step that is skipped on overflow.
- **`experiments/r2_enwik8/`:**
  - `run.py`, which mirrors ml-ladder's CLI and JSON fields (`memory`, `enwik8` and `copy` studies);
  - `parity.py`, the CPU check against ml-ladder's own `train()`;
  - `analyze.py`.
- **`tools/kaggle/`:**
  - the source travels as a private dataset (`git archive` of both repos);
  - kernels run one job per T4;
  - `push_r2.py` holds the presets.

  Kaggle unpacks uploaded tarballs, so the kernel finds each repo by a marker file.
- **A fix in `jaxexp.transformer`.** Explicit attention now accumulates its scores in fp32 (`preferred_element_type`), and `run.py` defaults to fused attention, matching ml-ladder's SDPA. In fp16, the two paths now agree bit for bit on CPU. A test with sharp attention catches the old path at 2.2%.

# Results

## Parity on CPU, before any GPU

12 configurations (GRU, LSTM, learned-position and RoPE transformers; seeds 0–2), 60 steps each, ml-ladder's `train()` against `jaxexp.train` from the same weights and batches, fp32:

- the training loss agrees to $5.9 \times 10^{-7}$ relative, at every step;
- the final parameters agree to $1.8 \times 10^{-6}$;
- validation bits/byte agrees to 6–7 significant digits.

## Calibration on the T4

| 25M RoPE step, T=1024, batch 16 | XLA temp memory |
|---|---|
| fp32, explicit attention | 10.13 GiB |
| fp16, explicit (fp16 scores) | 6.85 GiB |
| fp16, fused (fp32 scores) | 9.00 GiB |
| fp16, explicit, per-block remat | 2.28 GiB |

XLA's *CPU* compiler had put the fp32 step at 19.7 GiB (M1, O1). On the GPU the same program needs about half that. The remat that O1 judged necessary is not needed, so R2 runs without it.

## R2: every arm lands on ml-ladder's number

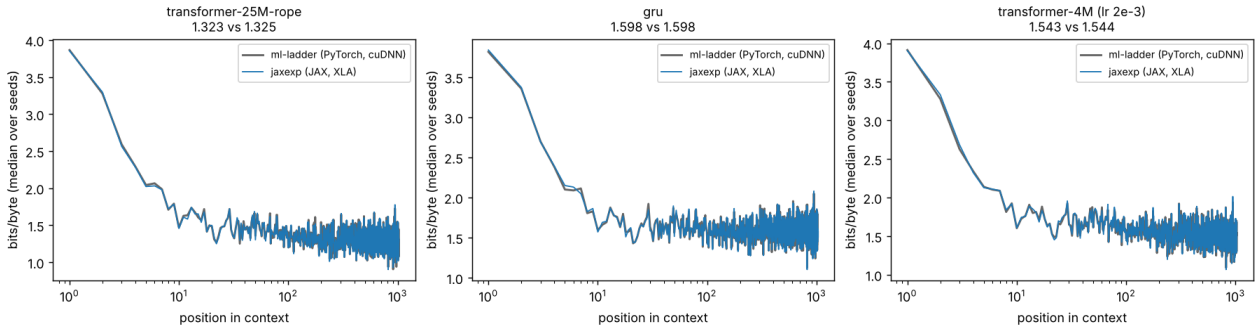



**Figure 1.** Loss by position, median over seeds 0-2, against ml-ladder's recorded curves

| arm (12,000 steps, T=1024) | ml-ladder bits/byte | JAX bits/byte | per-seed difference (s0, s1, s2) |
|---|---|---|---|
| 25M RoPE, lr 6e-4 | 1.3251 (1.3220–1.3320) | **1.3226** (1.3215–1.3322) | +0.0002, −0.0006, −0.0024 |
| GRU 2×592, lr 2e-3 | 1.5982 (1.5972–1.5987) | **1.5979** (1.5966–1.5989) | +0.0007, +0.0002, −0.0016 |
| 4M, lr 2e-3 | 1.5437 (1.5385–1.5547) | **1.5435** (1.5388–1.5585) | +0.0038, −0.0002, +0.0003 |
| *25M RoPE, explicit, fp16 scores* | | *1.3250 (1.3234–1.3339)* | *+0.0019, +0.0014, −0.0001* |

| arm (12,000 steps, T=1024) | ml-ladder bits/byte | JAX bits/byte | per-seed difference (s0, s1, s2) |
|---|---|---|---|
| *4M, explicit, fp16 scores* | | *1.5421 (1.5371–1.5601)* | *+0.0054, −0.0016, −0.0014* |

- **Every pair is within 0.006 bits/byte**, and 13 of 15 within 0.003. ml-ladder's own seed range is 0.010 for RoPE and 0.016 for the 4M. Each median lands within 0.0025 of ml-ladder's.
- **The curves agree position by position.** Averaged over positions 16–255, the medians differ by at most 0.004 bits/byte; over 256–1023, by at most 0.001. Single positions differ by up to 0.094, which is sampling noise: each position is scored on 256 windows.
- **fp16 was healthy.** At most 3 of 12,000 steps were skipped per run. The final loss scale was $2^{17}$ to $2^{22}$.
- **The fp16 score rounding is invisible here.** The explicit arms land inside the same ranges, so on enwik8 the rounding costs nothing measurable. It mattered on the copy task (below).

### Cost: XLA against PyTorch on the same T4

| arm | JAX tokens/s | PyTorch tokens/s | JAX peak memory | PyTorch peak | compile |
|---|---|---|---|---|---|
| 25M RoPE, fused | 34.4k | 43.9k | 9.9 GB | 3.2 GB | 16 s |
| 25M RoPE, explicit, fp16 scores | 46.8k | | 7.7 GB | | 15 s |
| GRU (`lax.scan` against cuDNN) | 80.7k | 179.6k | 1.2 GB | 0.7 GB | 3 s |
| 4M, fused | 121.9k | 231.4k | 3.1 GB | 0.9 GB | 7 s |
| 4M, explicit, fp16 scores | 161.2k | | 2.3 GB | | 7 s |

*Medians over 3 seeds. The PyTorch figures are ml-ladder's recorded runs on the same T4 type (fp16 autocast, SDPA, cuDNN).*

- **`lax.scan` runs at 0.45× cuDNN.** The scan issues one small matmul per step; cuDNN fuses the whole sequence.
- **XLA's attention materialises the T×T scores; SDPA on the T4 does not.** That accounts for the 3× memory gap. How much of the speed gap it explains is not measured. On the T4, `jax.nn.dot_product_attention` lowers to XLA's own implementation, not a flash kernel. JAX's cuDNN attention targets newer GPUs (not tried here), and Pallas doesn't run on a T4 (CLAUDE.md).
- **The explicit path with fp16 scores beats PyTorch** on the 25M model (46.8k against 43.9k): fp16 scores halve the score matrix's memory traffic. The fixed explicit path (fp32 scores) computes the same numbers as the fused one. Its GPU speed is not measured; expect the fused path's.

### R3: the copy task

| second-half accuracy | n=32 | n=128 | n=512 |
|---|---|---|---|
| LSTM, ml-ladder | 0.903 (0.865–0.912) | 0.031 (chance) | 0.031 |
| LSTM, JAX | 0.893 (0.864–0.910) | 0.031 | 0.031 |
| transformer 4M, ml-ladder | 1.000 | 1.000 | 0.999 (0.917–1.000) |
| transformer 4M, JAX, fused (Colab T4) | 1.000 | 1.000 | **1.000** (0.981–1.000) |
| *transformer 4M, JAX, explicit, fp16 scores* | *1.000* | *1.000* | *0.997 (0.273–1.000)* |

Both of ml-ladder's claims hold:

- **The transformer copies 512 symbols.** With fused attention, all three seeds do: 0.981, 1.000 and 1.000, against ml-ladder's 0.917, 1.000 and 0.999.
- **The LSTM fails at 128.** It reproduces per seed at n=32 (0.910 / 0.893 / 0.864, against 0.912 / 0.903 / 0.865) and is at chance from n=128 on.

**The one miss, and what it found.** In the first R3 kernel, the explicit-attention transformer (fp16 scores) reached only 0.273 on seed 1 at n=512, where ml-ladder has 1.000. Its loss was still on the plateau at step 750 (3.45, against 2.99 and 3.20 for seeds 0 and 2). Controls on that one configuration:

| seed 1, n=512, 1,500 steps | runs | loss at step 750 | accuracy |
|---|---|---|---|
| fp16, explicit (fp16 scores) | 3 | 3.40–3.46 | 0.144, 0.273, 0.579 |
| fp32, explicit | 3 | 2.68–2.70 | 0.998–1.000 |
| fp16, fused (fp32 scores) | 3 | 2.70–2.85 | 1.000 ×3 |
| *seed 0, fp16, explicit (repeat)* | 1 | 2.72 | 1.000 |

- **XLA on the GPU is not run-to-run deterministic.** Identical runs from identical weights and batches agree to step 375, then diverge (0.14, 0.27, 0.58).
- **The lag is systematic anyway.** Three of three fp16-score runs stay on the plateau. Six of six runs with fp32 scores leave it by step 750. So does the seventh, R3's fused rerun on Colab (2.68 at step 750, 1.000 at the end).
- **That is why ml-ladder's protocol choice (SDPA) matters, and why the explicit path was fixed.** Copying 512 symbols needs attention sharp enough that a rounding of about 0.02 in the scores slows the transition. With 1,500 steps, the transition sits right at the edge of the budget: ml-ladder's own seed 0 reached only 0.917.

## What surprised me

**The seed-for-seed agreement.** Two frameworks, two fp16 schemes, and two attention implementations give fifteen 12,000-step runs. Each lands within 0.006 bits/byte

of its PyTorch twin, and most within 0.002. Training on enwik8 at this scale is far less chaotic than the copy task's plateau. The same initial weights and batches give the same function, to three decimals.

**A precision bug that only one task could see.** Rounding attention scores to fp16 left enwik8 untouched: all six explicit runs are inside the reference ranges. On the copy task, seeds 0 and 2 still copied 512 symbols. But it reliably delayed seed 1's escape from the plateau. Nondeterminism made it look like noise at first. It took repeats of *both* precisions to see that one arm was 0/3 and the other 6/6.

**CPU memory numbers mislead about GPU programs.** XLA's CPU compiler put the 25M step at 19.7 GiB and implied remat was mandatory. The GPU compiler needs 10.1 GiB in fp32 and 6.85 GiB in fp16. M1 had flagged this to be confirmed on the T4, and it was the first thing the calibration kernel overturned.

**JAX loses on speed here, by a hardware accident.** The T4 is too old for cuDNN's fused attention and for Pallas, so XLA's attention materialises the score matrix. On sm_80+ the comparison would be different. M6's TPU work is where JAX's own kernels get a fair test.

## Open threads

- **The sm_80+ comparison.** Rerun the throughput rows where cuDNN flash attention and Pallas exist: an A100 or L4 elsewhere, or the TPU.
- **The copy task's transition time as a precision probe.** Steps-to-escape is a sensitive, cheap measure of attention numerics. Sweep the score precision (fp16, bf16, fp32) and n, with 5+ repeats each, given the nondeterminism.
- **Deterministic XLA on GPU.** `--xla_gpu_deterministic_ops` should make repeats identical. What it costs in speed is unmeasured.
- **A faster recurrence.** The GRU runs at 0.45× cuDNN. N4's `lax.associative_scan` applies only to linear recurrences, but an input-projection-hoisted scan with fewer, larger steps (chunking) is untried.